SPHOORTHY ENGINEERING COLLEGE | *(UGC Autonomous — NAAC Grade A)*

Kinematics & Dynamics of Machinery: Cams, Gears & Balancing

Department: MECHANICAL	**Course / Subject:** Kinematics & Dynamics of Machinery (ME501PC)
Semester: Semester 5	**Regulation:** Autonomous (SR23 / SR24)

Course Overview & Key Focus: Four-bar linkages, Inversions, Relative velocity method, Cam displacement profiles (SHM, Cycloidal), Involute gear tooth action, Epicyclic gear trains, and Balancing.

Unit 1: Mechanisms & Inversions

Kinematic links, pairs, kinematic chains, Grashof's law, Inversions of four-bar chain, Single slider crank inversions, Double slider crank inversions (Oldham's coupling).

Unit 2: Velocity & Acceleration Analysis

Relative velocity method, Instantaneous center of rotation (Kennedy's theorem), Coriolis acceleration component derivation in quick-return motion mechanisms.

Unit 3: Cam Profiles & Kinematics

Types of cams and followers, Follower motion curves (Uniform velocity, Simple Harmonic Motion SHM, Uniform acceleration, Cycloidal), Graphical profile construction.

Unit 4: Toothed Gearing & Gear Trains

Law of conjugate gearing, Involute and Cycloidal profiles, Interference and Undercutting prevention, Minimum number of teeth, Epicyclic gear train velocity ratios.

Unit 5: Balancing & Flywheels

Static and dynamic balancing of rotating masses in single/multiple planes, Turning moment diagrams of engines, Fluctuation of energy, Flywheel design principles.